Global land analysis & discovery
Investigating the impact of global land surface changes from a laboratory at the University of Maryland. **http://dgit.in/mar21-19**

Monitoring forests
Global Forest Watch allows you to see new tropical deforestation, and climate change.
http://dgit.in/mar21-20

tion is most likely to have occurred. As the system tracks an area over time, it can monitor changes continuously, flagging potential deforestation events, confirming them if the trend continues, and then going back and identifying the time when the changes first began. Satelligence goes from raw satellite data to pixel level detection, and it processes hundreds of thousands of images from multiple satellites to gather the intelligence. Satelligence goes a step further, and allows their stakeholders to prioritise the deforestation alerts based on various factors, such as how extensive the deforestation is, or if the deforestation is within a protected area. Satelligence counts major FMCG giants around the world as part of its clients including Unilever, Mondelez and Nestle. The World Bank and WWF are also customers.

The rainforests in South America are a critical carbon sink for the planet, and in Brazil the tropical jungles once covered an area as large as half of continental Europe. Deforestation in the Amazon peaked in the early 2000s, but then began to drop after 2005. Between 2007 and 2011, a drop in deforestation has been linked to the implementation of a satellite based alerting system called Real Time Deforestation Detection System or DETER (don't ask how that acronym works), that used to track incidences of deforestation and then fine the parties involved. The satellite based system captured and processed georeferenced imagery every 15 days, after which a government body would channel the alerts to local law enforcement authorities. Research into the efficacy of DETER has shown that an increase in the number of fines in a particular year, is very effective at reducing the amount of deforestation that occurs the following year. It is estimated that from the period between 2007 and 2011 alone, deforestation in the Amazon rainforest was reduced by a staggering 75 percent, because of satellite based alerts.

EYE IN THE SKY

While satellites are great, the ideal platform for monitoring forests are planes or drones, because of how close they can get to the subject of the study. Simply flying the plane at different altitudes allows researchers to fine tune the area they want to study, and in how much detail they want to study it. The Global Airborne Observatory (GAO), previously known as Carnegie Airborne Observatory (CAO) is a plane kitted out with the most cutting edge sensors, capable of observing and investigating forests in a number of ways. On board is a LIDAR and a powerful spectrometer, which together can be used to derive all sorts of information about a forest, from how much carbon it is capable of storing, which are the species growing in which areas, whether the trees are growing fast or slowly, the

Credit: MAAP

Deforestation in Peru because of a papaya plantation, captured on drone

chemical contents of the forest, and the various signatures of biochemical processes... all from the air. This information is then combined using algorithms into high resolution three dimensional maps. The technology has been deployed all around the world, and helps in conservation efforts as far apart as Belize and Borneo. As a consequence of investigating which are the forest regions that work the best as carbon sinks, and which are the areas that are most valuable or suitable for conservation, it was necessary to first understand how much carbon the forests were capable of capturing, and how valuable they were as carbon sinks. Such studies can help policymakers make more informed decisions on land use and conservation efforts. A surprising finding of these investigations was that deforestation was actually a strong source of carbon emissions and were 24.3 percent higher than previous assessments. The technology is increasingly being integrated into satellites and other airborne platforms.

When it comes to conservation, one of the most important factors is getting the support of the indigenous locals. The communities that live deep in the remote regions of the Amazon, have the best knowledge of their environment and surroundings, and are also deeply invested in ensuring the health of the ecosystem. It is with the cooperation of the locals and law enforcement officials that incidences of illegal deforestation, mining and agriculture can be identified and curtailed. To help the locals protect their forests better, organisations such as Amazon Conservation and WWF are providing them with drones, and the necessary training to use them. The drones allow the locals to quickly find illegal activities, and alert the law enforcement authorities in time. The locals are also using other sophisticated technologies, including Google maps, remote sensing data and GPS devices to monitor and track land use in the area.

The crucial factor for the success of the use of monitoring technologies for reducing deforestation, is the timeliness of the alerts. Often these pings allow law enforcement agencies to capture the perpetrators red handed, while still in the act of illegal logging, agriculture, mining or grazing. What makes these systems even more powerful, is pairing the technologies with the indigenous knowledge of the locals.